Formalised Treatment in the Symptom Network

We formalise the effect of cognitive behavioral therapy (CBT) treatment on the major depressive disorder (MDD) symptom network (Borsboom, 2017; Cramer et al., 2016) based on the common Dutch treatment manual *protocollaire behandeling van patienten met een depressieve stoornis* within the *Protocollaire behandeling voor volwassen met psychische klachten* (Keijsers, 2017) and the Ising model inspired formalisation of the MDD symptom network as a complex system (Cramer et al., 2016). To increase understandability and based on the model assumptions defined in Appendix A the model is divided into (1) a *Patient Layer*, (2) a *Treatment Layer*, and (3) a *Therapist Layer*, where each layer holds its own functions, and can selectively observe or interact with another layer. For instance, the *Therapist Layer* can activate interventions on the *Treatment Layer*, which then act on the *Patient Layer*.

The Patient Layer

The *Patient Layer* describes the patient in terms of the MDD symptom network based on the discrete time Cramer et al. (2016) formalisation, a time series of symptom activation $\mathcal{M}_i(t)$ and coactivation states $\mathcal{M}_{ij}(t)$, and a symptom relationship intuition $\mathcal{R}(t)$. We assume that the MDD symptom network can be studied independently and consists of $J = 9$ symptoms, corresponding exactly to the DSM-5 criteria for MDD (American Psychiatric Association, 2022):

$$V = \{DM, IP, AW, IH, PAR, FE, WG, CTI, S\}$$

Cramer et al. (2016) define each symptom $i \in V$, at each time step $t \in \{0,1, \dots, T\}$ – the number of days since treatment was started – as either present or absent, $X_i(t) \in \{0, 1\}$. Each symptom i has a threshold $b_i(t) \in \mathbb{R}_{>0}$ and is connected to other symptoms j by weights $W_{ij}(t) \in \mathbb{R}_{>0}$. Weights are assumed to be symmetric, $W_{ij}(t) = W_{ji}(t)$. We understand $b_i(t)$ as

an approximation of symptom specific mental processes biasing its tendency to be present – also accounting for autocorrelating or self-reinforcing effects (hence., $W_{ii}(t) = 0$) – representing the external field τ in the traditional Ising model. $W_{ij}(t)$ is understood as approximating mental causal processes between symptoms corresponding to the coupling in the traditional Ising model. The set of all undirected connections in the network is $E = \{\{i, j\} : i, j \in V, i \neq j\}$.

The probability that symptom i is present at time t depends on which of its neighbors were present at $t - 1$, weighted by connection strength W_{ij} (activation function)

$$A_i(t) = \sum_{j=1}^{J} W_{ij}(t-1)X_j(t-1)$$

and the gap between its activation and its threshold (probability function)

$$P(X_i(t) = 1) = \frac{1}{1+e^{b_i(t)-A_i(t)}}.$$

Note that contrary to the Cramer et al. (2016) model, we define $W_{ij}(t-1)$ and $b_i(t)$ dynamically. Also note that the inverse temperature β, as defined in the Ising model, is constant.

Keijsers (2017) bases treatment progress and treatment targets on patient recollection and introspection. Since internal states, such as, thresholds $b_i(t)$ and weights $W_{ij}(t)$ cannot be observed directly, we suggest they are reconstructed by the patient from observable symptom activation states. So far, we found some preliminary research suggesting an extensive tradition of assuming that memory and introspection both may be reconstructed supporting this idea – yet an extensive review of the literature still needs to follow (Bartlett, 1995; Schacter, 2012). Hence, we propose the *Patient Layer* can communicate introspected internal states through their memory $\mathcal{M}_i(t)$ of symptom activation frequency and intuition $\mathcal{R}(t)$ of symptom relations. Both $\mathcal{M}_i(t)$

and $\mathcal{R}(t)$ are deliberately defined as simplistic frequency heuristics in the current simulation. More elaborate models exist and may be extended upon in the future (e.g., Bayesian models of memory reconstruction; Hemmer & Steyvers, 2009; Xu, Hemmer, & Zhang, 2025).

Both memory $\mathcal{M}_i(t)$ and intuition $\mathcal{R}(t)$ are thought to cover the preceding two weeks (i.e., fourteen timesteps t), allowing both temporal dynamics to be captured while also providing a plausible range of retrospection on symptom frequency, $\mathcal{R}(t)$. We define, thus, the memory $\mathcal{M}_i(t)$ as

$$\mathcal{M}_i(t) = \sum_{u=t-N+1}^{t} X_i(u),$$

the biweekly frequency of symptom i at time t. Additionally, we define

$$\mathcal{M}_{ij}(t) = \sum_{u=t-N+1}^{t} X_i(u)X_j(u) \in \{0, \dots, 14\}, \quad \mathcal{M}_{ij} \leq \min(\mathcal{M}_i, \mathcal{M}_j),$$

as the biweekly frequency of symptom i and symptom j concurring at time t. We propose intuition $\mathcal{R}(t)$ of symptom co-occurrence of all symptom pairs based on symptom frequency. In $\mathcal{R}(t)$ symptom pairs that are consistently simultaneously active $\{i, j\}$ should be included, while those that are rarely simultaneously active should be excluded. Hence, we define

$$d_{ij}(t) = \frac{\mathcal{M}_{ij}(t)}{\mathcal{M}_i(t) + \mathcal{M}_j(t) - \mathcal{M}_{ij}(t)} \in [0,1], \quad d_{ij} \coloneqq 0 \text{ if } \mathcal{M}_i(t) = \mathcal{M}_j(t) = 0,$$

where $d_{ij}(t) \to 1$ the more symptom i and symptom j co-occur simultaneously and $d_{ij}(t) \to 0$ the more i and j activate independently from one another (Jaccard, 1912; Tanimoto, 1958). We include a symptom pair $\{i, j\}$ into the intuition set $\mathcal{R}(t)$ that shows more co-activation than independent activation

$$\mathcal{R}(t) = \left\{\{i, j\} : d_{ij}(t) \geq \kappa\right\}, \quad \kappa = 0.5,$$

where κ serves as an inclusion cut-off, establishing that symptoms must co-occur at least half as often as they appear in total. The robustness of the current value of κ is still to be determined with alternative values, for instance $\kappa \in \{0.4,0.5,0.6\}$.

The Treatment Layer

The *Treatment Layer* holds formalised interventions based on the *protocollaire behandeling van patienten met een depressieve stoornis* within the *Protocollaire behandeling voor volwassen met psychische klachten* (Keijsers, 2017). We assume the following: First, each intervention successfully and directly addresses underlying processes captured in specific network parameters $W_{ij}(t)$ and $b_i(t)$ while it is performed. Second, the extent to which an intervention modifies these parameters is bounded by the specific underlying processes that are addressed. Third, the extent to which an intervention modifies the parameters at any given time depends on the ability of the patient to perform that specific intervention successfully.

Hence, we define an intervention I_k, in the set of all K interventions $\mathcal{I} = \{I_1, \dots, I_K\}$, as

$$I_k = (\lambda_k, \gamma_k, t_k),$$

where λ_k and γ_k modify network parameters, and t_k holds the first onset of intervention k. Network parameters of symptoms $\mathcal{S}_k \subseteq V$ and connections $\mathcal{E}_k \subseteq E$ are modified in

$$\lambda_k : \mathcal{S}_k \to M \text{ and } \gamma_k : \mathcal{E}_k \to M$$

by a set of multiplier values $M = \{0.8, 1.1\}$ based on Cramer et al.'s (2016) global multiplier, where $m = 0.8, m \in M$ denotes a decrease and $m = 1.1$ denotes an increase. Note that these are not symmetrical – a single intervention that increases a parameter value cannot compensate for a

single previous decrease in that parameter, as $0.8 * 1.1 \neq 1$. λ_k and γ_k capture the maximal parameter change a single intervention I_k can induce.

The extent of change in target parameters during intervention I_k at time t depends on the patient's ability to successfully implement the intervention at that time. We call this ability competence $q_k(t)$. Skill acquisition is classically described by exponential learning curves, in which competence rises quickly and then decelerates toward mastery (Pusic et al., 2015; Donner & Hardy, 2015). We take this as our point of departure but adapt it to the success-based learning the manual emphasises: Keijsers (2017) prioritises feasibility and the avoidance of failure, we assume competence accrues only through successful implementations. Gating the exponential curve by the current success rate (Leibowitz et al., 2010) transforms it into a sigmoidal trajectory, where competence grows slowly while few successes have accumulated, accelerates as successes compound and saturates as it approaches mastery. Thus, we define competence, $q_k(t) \in (0,1)$, as the success rate of the patient in implementing intervention I_k at time t. Consequently,

$$q_k(t) = q_k(\infty) - \big(q_k(\infty) - q_k(t-1)\big)\, e^{-\alpha_k\, q_k(t-1)},$$

where $q_k(\infty)$ is the maximum achievable competence, $q_k(t-1)$ is the success rate at the previous time point $t-1$, and $\alpha_k \in \mathbb{R}_{>0}$ specifies the learning rate of intervention I_k (derived in *Therapist Layer*. For low levels of the success rate $q_k(t-1)$ increases in $q_k(t)$ will be slow, accelerating exponentially, while for high levels of $q_k(t-1)$ competence $q_k(t)$ approaches the asymptotic ceiling $q_k(\infty)$. For all I_k we assume a low initial level of competence, $q_k(t_k) = 0.1$, and mastery, $q_k(\infty) = 1$, simplifying to

$$q_k(t) = 1 - (1 - q_k(t-1))\, e^{-\alpha_k\, q_k(t-1)}.$$

Since competence is understood to increase due to successful implementation of the intervention, it should only increase while an intervention is active and held constant otherwise. Thus, we define competence $q_k(t)$ formally as

$$q_k(t) = \begin{cases} 1 - (1 - q_k(t-1))\, e^{-\alpha_k\, q_k(t-1)} & \text{if } I_k \text{ is active at } t \\ q_k(t-1) & \text{if } I_k \text{ is inactive at } t \text{'} \end{cases}$$

where I_k is activation at time t is determined by the *Therapist Layer*.

Each intervention I_k rescales its targeted parameters multiplicatively from baseline toward the maximal change set by λ_k and γ_k, governed by their competence $q_k(t)$. Two properties follow: First, at competence $q_k(t) = 0$ any intervention leaves the parameter unchanged, and while at mastery $q_k(\infty) = 1$ the full multiplier λ_k is applied. Second, because each factor is strictly positive, parameters will remain positive with any number of interventions I. Third, the maximal change an intervention I_k induces anchors to the value the parameter holds at onset of the intervention, including the effect of already active parameters. Parameters $b_i(t)$ and $W_{ij}(t)$ in the *Patient Layer* at time t are, hence, defined by

$$b_i(t) = b_i(0) \prod_{k:i \in T_k} \lambda_k(i)^{q_k(t)} \text{ , and}$$

$$W_{ij}(t) = W_{ij}(0) \prod_{k:\{i,j\} \in E_k} \gamma_k(\{i,j\})^{q_k(t)} \text{ ,}$$

where $b_i(0)$ and $W_{ij}(0)$ are the disordered baseline parameters produced by I_0 from the healthy network (see *Deriving a Theory-Based Disordered Symptom Network*). The specific interventions I_k are theoretically based on *Lewinsohn's Behavioral Activation* (LBA), *Kanter's Behavioral Activation* (KBA), or *Beck's Cognitive Therapy* (CT) as described in Keijsers (2017) and derived in Table 1.

Implicitly, thus, it is assumed (1) that ability to perform an intervention is equivalent to the change the intervention can produce and (2) that at mastery, $q_k(\infty) = 1$, the underlying psychological mechanism is addressed in its entirety and (3) that when several interventions target the same parameter, they address distinct underlying processes, so that their proportional effects combine multiplicatively.

The Therapist Layer

The *Therapist Layer* describes the therapist and holds formalised decision rules following the structure of the four treatment phases described in Keijsers (2017) determining which interventions $I_k \in \mathcal{I}$ are active at time t. Following the treatment manual (Keijsers, 2017) we assume the following: (1) Therapy occurs once a week (i.e., every seven time steps); (2) The therapist observes the patient's symptom level as reported through self-assessment; (3) The therapist observes the patient's symptom associations through their interaction – for instance, through Socratic questioning; (4) The therapist follows each of the four described treatment phases exactly as described; (5) The therapists determines the onset of the treatment phases and their variants based on either their natural sequence or the symptom level; (6) The therapist determines treatment targets based on the patient's reported symptom associations and symptom levels.

We define the set of therapy days

$$\mathcal{T}_{therapy} = \{\theta_n : n = 0, \dots, N_S\} = \{0,7,14, \dots\},$$

where n is the therapy session number and $\theta_n = 7n$ denotes the timestep t at which session n occurs. During each therapy day θ_n, the *Therapist Layer* receives both memory $\mathcal{M}_i(\theta_n)$ as well as intuition $\mathcal{R}(\theta_n)$ from the *Patient Layer*, which, based on decision rules, determine the active

interventions I_k from the *Treatment Layer* at time t. As suggested in Keijsers (2017) MDD symptom severity is assessed each session n using a self-report questionnaire. Instead of the indicated BDI, we formalise assessment using the PHQ-9 a common alternative estimating MDD symptom frequency (Kroenke, Spitzer, & Williams, 2001). PHQ-9 items are rated on a 4-point Likert scale (0 = *not at all*; 1 = *several days*; 2 = *more than half the days*; 3 = *nearly every day*) and correspond exactly to the nine DSM symptoms in V. We define the PHQ-9 score for symptom i during session n based on the preceding two weeks, thus, as

$$\mathrm{PHQ}_i(\theta_n) = \begin{cases} 0 & \mathcal{M}_i(\theta_n) = 0 \\ 1 & 1 \leq \mathcal{M}_i(\theta_n) \leq 6 \\ 2 & 7 \leq \mathcal{M}_i(\theta_n) \leq 10 \\ 3 & 11 \leq \mathcal{M}_i(\theta_n) \leq 14 \end{cases},$$

and the total $\mathrm{PHQ}(\theta_n) = \sum_{i \in V} \mathrm{PHQ}_i(\theta_n) \in \{0, \dots, 27\}$. Traditional cut-offs are used to define depression severity

$$\sigma(\theta_n) = \begin{cases} 0\ (\text{none}) & 0 \leq \mathrm{PHQ}(\theta_n) \leq 4 \\ 1\ (\text{mild}) & 5 \leq \mathrm{PHQ}(\theta_n) \leq 9 \\ 2\ (\text{moderate}) & 10 \leq \mathrm{PHQ}(\theta_n) \leq 14, \\ 3\ (\text{mod. severe}) & 15 \leq \mathrm{PHQ}(\theta_n) \leq 19 \\ 4\ (\text{severe}) & 20 \leq \mathrm{PHQ}(\theta_n) \leq 27 \end{cases}$$

at therapy session n.

Following Keijsers (2017), we define four treatment phases F that, dependent on treatment progress (i.e., $\mathcal{M}_i(\theta_n)$ and $\mathcal{R}(\theta_n)$ reduction), span sessions $F_1 = \{n : 1 \leq n \leq 12\}$, $F_2 = \{n : 3 \leq n \leq 16\}$, $F_3 = \{n : 6 \leq n \leq 18\}$, and $F_4 = \{n : 16 \leq n \leq 20\}$, reflective of common individual treatment development.

Phase 1: Behavioral Activation

Treatment begins with *Lewinsohn's Behavioral Activation* (LBA; Keijsers, 2017) corresponding to interventions I_1, I_2, I_3, and I_4. At session 1 intervention I_1 is implemented and the patient begins to register their mood in relation to whether they are performing pleasurable or necessary activities. Based on these registrations, then, session 2 starts with intervention I_2, where the patient maximises the registered pleasurable activities and minimises the registered necessary activities. Assuming that difficulties to expand activities may occur, in session 3, problem solving abilities are improved and avoidance behavior is addressed through intervention I_3 and I_4.

Phase one is continued as long as the patient is responding to treatment. Treatment response is understood as a reduction of MDD symptom severity indicated by PHQ-9 scores. We define treatment response loosely as a decrease in the total $\mathrm{PHQ}(\theta_n)$ in either of the last to transitions,

$$\mathrm{PHQ}(\theta_n) < \mathrm{PHQ}(\theta_{n-1}) \vee \mathrm{PHQ}(\theta_{n-1}) < \mathrm{PHQ}(\theta_{n-2}),$$

accounting for incidental score increases without prematurely abandoning LBA as an effective intervention. A transition to phase two follows either after the successful implementation of LBA, which we define as a qualitative drop in depression severity

$$\sigma(\theta_n) < \sigma(\theta_{n-1}),$$

where we assume that a change in $\sigma(\theta_n)$ represents a clinically meaningful shift, or by the unsuccessful implementation of LBA

$$\mathrm{PHQ}(\theta_n) \geq \mathrm{PHQ}(\theta_{n-1}) \geq \mathrm{PHQ}(\theta_{n-2}) \quad \text{or} \quad n = 12,$$

where $n = 12$ is the session index capping phase one to its maximal length in F_1.

Phase 2: Evaluate Behavioral Activation and Investigation of Automatic Negative Thoughts

Keijsers (2017) describes that the second phase of treatment starts with the evaluation of LBA. Accordingly, treatment progression is determined by the evaluation of LBA leading to the transition from phase one to phase two.

If LBA is deemed successful, the first session of phase two starts immediately with Beck's Cognitive Therapy (CT). During intervention I_7, the patient together with the therapist investigates their negative automatic thoughts (also called *hot cognitions*) and attempts to challenge their credibility. Continuously, based on $\mathcal{M}_i(\theta_n)$, the hot cognition $h \in H$ with the highest frequency is selected and intervened upon. We approximate hot cognitions with the hot cognition set $H = \{DM, WG, S\}$, which includes all high emotionality symptoms thought to approximate hot cognitions at the symptom level. Although the target hot cognition h may change each session, the competence $q_7(t)$ is shared between hot cognitions H throughout intervention I_7, representing increased skill in challenging negative automatic thoughts.

If LBA is deemed unsuccessful, in the first session of phase two, additionally to starting with CT, LBA will be disregarded (i.e., I_1, I_2, I_3, and I_4 become inactive). Instead, Kanter's Behavioral Activation (KBA) is introduced. The patient starts with intervention I_5, creating an extensive list of life goals in various life domains and coming up with an activity hierarchy that is used to advance towards these life goals. Since difficulties, again, may occur in implementing the intervention I_5, during intervention I_6 in the next session, functional analysis is used to address these problems through a variety of techniques.

Phase two is continued for at least three sessions. We derive this from Keijsers (2017) suggested phase range – since the earliest start of phase three is in session six and the earliest

start of phase two is in session three, phase two must span at least three sessions. Since Keijsers (2017) additionally specifies that a change to phase three follows once a patient has learned to examine automatic thoughts (i.e., phase three is contingent on competence $q_7(\theta_n)$, we use this to specify the α within competence $q_k(t)$. Specifically, we define competence $q_7(\theta_n) \geq 0.5$ as meeting the competence criterion. Based on $q_k(t_k) = 0.1$, to meet the competence criterion, we set $\alpha_k = \alpha = 0.15$, which if I_7 starts in session three, surpasses the criterion at approximately session 5.3. Accordingly, since the criterion is only evaluated at sessions, a transition to phase three occurs if $q_k(\theta_n) \geq 0.5$. To test for robustness, other values of α will be considered (e.g., $\alpha = 0.10$, $\alpha = 0.20$).

Phase 3: Avoidance Breakthrough and Investigation of Assumptions and Core Beliefs

The respective behavioral activation – LBA or KBA – as well as challenging of negative automatic thoughts are continued in phase three. Additionally, the therapist introduces the notion of schemas – larger latent maladaptive thinking patterns based on core beliefs or underlying assumptions. To identify schemas, the *downward arrow technique* is commonly used, which uses previously found negative automatic thoughts as a starting point and uses specific questions to identify the underlying core beliefs and assumptions. By investigating a variety of thoughts, common core beliefs and assumptions shared across thoughts can be identified. Similarly to the challenging of hot cognitions, the credibility of schemas is then challenged.

Formally, the therapist, again, chooses the most frequent hot cognition symptom h with the highest frequency based on $\mathcal{M}_i(\theta_n)$ and assigns it to a schema cluster C. Then, all symptoms j contained in the intuition set $\mathcal{R}(\theta_n)$ identified by the patient to co-occur with j are added to schema cluster

$$C = \{h\} \cup \{j : \{h, j\} \in \mathcal{R}(\theta_n)\}.$$

Intervention I_8 then acts on all connections among its cluster members

$$\mathcal{E}_8 = \big\{\{i, j\} : i, j \in C, i \neq j\big\},$$

weakening the within-schema associations. As with challenging of negative automatic thoughts, while symptoms $i, j \in C$ may change each session, competence $q_8(t)$ is shared across targets.

Based on Keijsers (2017), phase three will be continued for ten sessions.

Phase 4: Relapse Prevention

In phase four the patient in collaboration with the therapist creates a relapse prevention plan in which they recap the interventions that were used during therapy, as well as common negative automatic thoughts and schemas that occurred during treatment. This prepares the patient to independently apply the techniques they learned to prevent the development of another MDD episode. Hence, we assume that the relapse prevention enables the respective interventions I_k covered in treatment to remain active

Deriving a Theory Based Disordered Symptom Network

To obtain disordered symptom networks, the initial network parameters W_{ij} and b_i modified based on the theoretical assumptions of MDD development and maintanence as described in the *protocollaire behandeling van patienten met een depressieve stoornis* within the *Protocollaire behandeling voor volwassen met psychische klachten* (Keijsers, 2017). Three theoretical perspectives are proposed: *Lewinsohn's Behavioral Activation* (LBA) perspective, defining depression as an absence of behavior coming from negative/a lack of positive reinforcement, *Kanter's Behavioral Activation* (KBA), defining depression as avoidance of

negative experiences leading to withdrawal, and *Beck's Cognitive Therapy* (CT), defining depression as maladaptive thoughts and schemas coming from negative automatic thoughts.

Based on the previously defined modifier functions

$$\lambda_k : T_k \to M \text{ and } \gamma_k : E_k \to M,$$

where $m = 0.8$ denotes a decrease and $m = 1.1$ denotes an increase, we formalise the parameter modification I_0 of a disordered MDD symptom network as derived in Table 2. Unlike the therapeutic interventions $I_1, \dots, I_K \in \mathcal{I}$, the disorder-inducing modification $I_0 \notin \mathcal{I}$ is not competence-gated but is applied once in full at time $t = 0$. Let $b_{i,\text{health}}$ and $W_{ij,\text{health}}$ denote the parameters of a healthy MDD symptom network. Applying I_0 produces the disordered baseline that enters the *Patient Layer* at $t = 0$,

$$b_i(0) = b_{i,\text{health}} \lambda_0(i) \text{ and } W_{ij}(0) = W_{ij,\text{health}} \gamma_0(\{i,j\}),$$

with $\lambda_0(i) = 1$ and $\gamma_0(\{i,j\}) = 1$ wherever Table 2 specifies no modification. All subsequent treatment rescales the disordered baselines as defined via

$$b_i(t) = b_i(0) \prod_{k : i \in T_k} \lambda_k(i)^{q_k(t)},$$

and the analogous

$$W_{ij}(t) = W_{ij}(0) \prod_{k : \{i,j\} \in E_k} \gamma_k(\{i,j\})^{q_k(t)}.$$

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Tables and Figures

Table 1

Treatment Manual-Based Formalisations of Interventions I_k

Page (Treatment)	k	Description	I_k		Rationale
			λ_k	γ_k	
p. 29 (LBA)	1	Registering of mood while performing activities that are either pleasurable or necessary to increase awareness between behavior and mood.	–	$\gamma_1(\{IP, DM\}) = 0.8$ $\gamma_1(\{IP, CTI\}) = 1.1$ $\gamma_1(\{CTI, DM\}) = 1.1$	We include symptoms related to mood (DM), experiencing pleasure (IP), and awareness (CTI). We assume that awareness of a connection between two symptoms (DM and IP) reduces their direct pairing, instead making it dependent on cognitive abilities (CTI).
pp. 29-30 (LBA)	2	Mood independent structured maximisation/minimisation of feasible	$\lambda_2(IP) = 1.1$	–	We include symptoms related to experiencing pleasure (IP). Since pleasurable experiences are engaged in independent of mood

Page (Treatment)	k	Description	I_k		Rationale
			λ_k	γ_k	
		pleasurable/necessary activities.			(DM), we assume that the threshold of IP is increased, to prevent activation even with strong DM connections.
p. 31 LBA	3	Increasing problem-solving abilities to address emotional avoidance and execution difficulties.	$\lambda_3(CTI) = 1.1$	–	We include symptoms related to emotional avoidance (IP), execution difficulties and problem solving (CTI). We assume that problem solving increases CTI ability, therefore, reducing engagement in IP.
p. 31 LBA	4	Raising awareness of emotional avoidance to the patient and addressing it directly.	$\lambda_4(IP) = 1.1$	$\gamma_4(\{IP, CTI\}) = 1.1$	We include symptoms related to emotional avoidance (IP) and awareness (CTI). We assume that awareness of a symptom (IP) increases its dependence on cognitive abilities (CTI). Directly addressing a

Page (Treatment)	k	Description	I_k		Rationale
			λ_k	γ_k	
					symptom (IP) is assumed to make its activation less likely.
pp. 32-33 (KBA)	5	Engaging in daily activities that advance behaviour-based (rather than achievement-based) life goals across a wide range of life areas, guided by an activity hierarchy.	$\lambda_5(WG) = 1.1$	–	We include symptoms related to self-efficacy/self-concept (WG). We assume that advancement in a wide variety of life areas increases an individual's self-efficacy/self-concept. This reduces the tendency of a person to feel worthless (WG).
p. 33 (KBA)	6	Implementation of stimulus control, skills training, or contracts and behavioral avoidance psychoeducation to address situation-, behaviour-, or consequence-based problems respectively, based on a functional analysis (FA).	$\lambda_6(CTI) = 1.1$	–	We include symptoms related to skills training (decision-making and problem-solving; CTI), and behavioral avoidance (CTI). Stimulus control acts on environmental factors and falls outside the symptom network's scope.

Page (Treatment)	k	Description	I_k		Rationale
			λ_k	γ_k	
					We assume that skills training helps adequate decision making based on learned strategies aiding decision making (CTI), and avoidance psychoeducation improves decision making, thereby, its impact on other symptoms, such as DM, IP, or FE.
pp. 36-38 (CT)	7	Challenging of negative automatic thoughts by evaluating their credibility against alternative thoughts.	$\lambda_7(h) = 1.1$	–	We include the therapist selected hot cognition symptom $h \in H = \{DM, WG, S\}$, which is thought to approximate hot cognitions (i.e., highly emotional thoughts) at the symptom level. We assume that challenging reduces the rate of hot cognitions and, thus, the activity of symptoms.
pp. 38-42 (CT)	8	Challenging of schemas (rules of living and assumptions) by evaluating	–	$\gamma_8(\{i,j\}) = 0.8$ for all $\{i,j\} \in \mathcal{E}_8$	We include the therapists selected schema $C \subseteq V$.

Page (Treatment)	k	Description	I_k		Rationale
			λ_k	γ_k	
		their credibility against alternative schemas.		$\mathcal{E}_8 = \{\{i,j\} : i,j \in C, i \neq j\}$	We assume that challenging of schemas reduces the automatic association between its symptoms.

Note. Interventions I_k are listed in the order presented in the *protocollaire behandeling van patienten met een depressieve stoornis* within the *Protocollaire behandeling voor volwassen met psychische klachten* (Keijsers, 2017). *Page* refers to where the intervention is described in the manual; *treatment* to the treatment type – Lewinsohn's Behavioral Activation (LBA), Kanter's Behavioral Activation (KBA), and Cognitive Therapy (CT); k indexes the intervention; *description* summarises the intervention as presented in Keijsers (2017); I_k gives the formalised parameter changes via λ_k and γ_k, where a value of 0.8 indicates a decrease and 1.1 an increase in the targeted parameter; and *rational* explains how I_k was derived.

Table 2

Formalised Initial MDD Symptom Network Parameters I_0

Page (Theory)	Description	I_0		Rationale
		λ_0	γ_0	
p. 22 (LBA)	Vicious cycle in which negative/lack of positive reinforcement leading to a reduction in patient activity level reduces the chance of again receiving positive reinforcement leading to *depressed behavior*.	$\lambda_0(FE) = 0.8$ $\lambda_0(PAR) = 0.8$ $\lambda_0(AW) = 0.8$ $\lambda_0(IH) = 0.8$	– – – –	We include symptoms related to activity levels (FE, PAR, AW, IH). We assume that activity levels are prone to reduce without external positive reinforcement, showing a tendency in FE, PAR, AW, and IH to be activated.
p. 22 (LBA)	Depressive behavior deterioration due to reactive feelings once enabling family support and accommodation falls away.	– – –	$\gamma_0(\{WG, FE\}) = 1.1$ $\gamma_0(\{WG, PAR\}) = 1.1$ $\gamma_0(\{WG, AW\}) = 1.1$	We include symptoms related to activity levels (i.e., depressive behavior; FE, PAR, AW, IH) and symptoms related to interpersonal feelings (WG). We assume that the deterioration of depressive behavior (FE, PAR, AW, IH),

Page (Theory)	Description	I_0		Rationale
		λ_0	γ_0	
				is a direct consequence of the interpersonal feelings (WG)
pp. 22-23 (KBA)	Avoidance behavior prevents unexpected negative emotional experiences thereby reinforcing withdrawal and preventing positive experiences.	$\lambda_0(CTI) = 0.8$	–	We include symptoms related effective decision making (CTI). IP is not included, since the patient is not anhedonic, but withdraws specifically because they experience emotions. We assume impaired decision making (CTI), leading to inappropriate comping strategies
p. 23 (CT)	Negative automatic thoughts (also, hot cognitions) lead to maladaptive thinking patterns.	$\lambda_0(WG) = 0.8$ $\lambda_0(DM) = 0.8$ $\lambda_0(S) = 0.8$		We approximate negative automatic thoughts at the symptoms level and include hot cognition symptoms, referring to symptoms which have high emotionality (WG, S, DM). We assume MDD patients are more prone to negative automatic thoughts.

Page (Theory)	Description	I_0		Rationale
		λ_0	γ_0	
p.23 (CT)	Maladaptive schemas are large latent thinking structures of assumptions and core beliefs that, once activated, may maintain depression.		$\gamma_0(\{i,j\}) = 2, i,j \in \mathcal{C}_{\text{generating}}$	We include all symptoms as potential schema participants. We assume that schemas are latent structures of high connectivity between symptoms i,j in schema $\mathcal{C}_{\text{generating}}$. These structures will lead to simultaneous activation of multiple symptoms once one of the schema symptoms is activated.

Note. Formalised modifications of healthy MDD symptom networks to obtain disordered MDD symptom networks based on LBA, KBA and CT theoretical assumptions in the *protocollaire behandeling van patienten met een depressieve stoornis* within the *Protocollaire behandeling voor volwassen met psychische klachten* (Keijsers, 2017). *Page* refers to the page where the formalised aspect can be found; $k = 0$ denotes that formalisations refer to the disordered symptom network setup; *description* summarises the formalised theoretical assumptions; I_0 gives the formalised parameter changes via λ_0 and γ_0, where a value of 0.8 indicates a decrease and 1.1 an increase in the targeted parameter; and *rational* explains how I_0 was derived.

* To define a local constructing schema cluster $\mathcal{C}_{\text{generating}}$, based on pre-existing connectivity patterns, we first select the symptom with the highest weights $a = \arg\max_{i \in V} \sum_{j \in V} W_{ij}$, $\mathcal{C} \leftarrow \{a\}$, which serves as anchor a from which symptoms will be added to the schema cluster $\mathcal{C}$. Then, the cluster grows outward in layers. At layer ℓ, each candidate symptom i not yet considered that is connected to a symptom j added in the previous layer is reached at distance $d_i = \ell$ and added to $\mathcal{C}_{\text{generating}}$ with probability $P(i \in \mathcal{C}) = 1 - e^{-W_{ij}/d_i}$, where W_{ij} is the weight of the traversed connection. Only added symptoms seed the next layer's expansion, and, thus, hop further from the anchor a ($d_i = \ell + 1$ relative to a layer-ℓ node). Each symptom $i \in V$ is considered only once as a candidate for $\mathcal{C}_{\text{generating}}$.

Appendix A. General Model Assumptions

(1) Major Depressive Disorder (MDD) is a symptom cluster that exists and can be simulated as well as studied in isolation from other symptoms.

(2) MDD phenomena can be simulated based on pairwise interactions between symptoms as in the Ising model or the simplified Cramer et al. (2016) model, which approximate causal interactions between symptoms as suggested by Borsboom (2017).

(3) MDD symptoms correspond exactly to the DSM-5 defined descriptions of symptoms.

(4) MDD symptoms can only be present (1) or absent (0).

(5) MDD symptoms have thresholds which approximate multiple processes within one symptom that bias its tendency to be present.

(6) MDD symptoms share weights (i.e., connections) which approximate multiple causal processes between pairs of symptoms in terms of bidirectional pairwise interaction.

(7) MDD symptom parameters (thresholds and weights) can only be positive

(8) Reasonable initial MDD parameter values correspond in size approximately to the parameters estimated by Cramer et al. (2016).

(9) The therapist can observe symptom activation (1) during a session and (2) indirectly through PHQ-9 reports

(10) The therapist starts and stops interventions based on currently observed symptom activation and previously observed symptom activation, as well as decision rules informed by the *Protocollaire behandling voor volwassen met psychische klachten* (Keijsers, 2017)

(11) Interventions directly influence specific parameter constellations

(12) Interventions perfectly influence specific parameter constellations

(13) Interventions continuously influence specific parameter constellations

(14) Interventions follow traditional exponential learning curves

(15) Interventions rely on success-based learning; thus, exponential learning curves are modified to follow a sigmoidal shape (Leibowitz et al., 2010)

(16) Interventions can be perfectly learned with sufficient time

(17) Interventions lead to parameter change proportional to learning of the specific treatment technique

(18) Each intervention induces parameter change bounded by a maximal change potential an intervention can produce

  (18.1) a reasonable increase in parameters (weights/thresholds) corresponds to a parameter change by a multiplicator of 1.1 (Cramer et al., 2016)

  (18.2) a reasonable decrease in parameters (weights/thresholds) corresponds to a parameter change by a multiplicator of 0.8 (Cramer et al., 2016)

(19) Interventions correspond exactly to what is specified within the *protocollaire behandeling van patienten met een depressieve stoornis* within the *Protocollaire behandeling voor volwassen met psychische klachten* (Keijsers, 2017)

Appendix B. Symptoms and their Abbreviations

Depressed Mood:	**DM**
Diminished Interest/Pleasure:	**IP**
Weight Change/Appetite Change:	**AW**
Insomnia/Hypersomnia:	**IH**
Psychomotor Agitation/Retardation:	**PAR**
Fatigue/Loss of Energy:	**FE**
Worthlessness/Inappropriate Guilt:	**WG**
Concentration/Thinking Problems/Indecisiveness:	**CTI**
Suicidal Ideation/Suicide Attempts/Suicide:	**S**

Note. Symptoms are exactly defined as provided in the DSM-5 under MDD. All interpretations of these symptoms are based on these specific definitions